

PRATYASHA BASAK

Email: prattub7@gmail.com
Github: [pratyasha7](#)

Ph-No: +91-6291525486
Linkedin: [PRATYASHA BASAK](#)

PROFILE SUMMARY

Software Engineer with a strong foundation in Computer Science, focused on building scalable backend systems, data pipelines, and mathematically-grounded ML applications. Experienced in end-to-end system design — from API-based data ingestion and abstraction layers to predictive algorithm implementation and model integration — with a strict focus on data integrity and system reliability. Proficient in Python, SQL, and modern software engineering practices including version control, modular architecture, and collaborative development. Proactive problem-solver with a track record of delivering measurable engineering impact in production-facing environments.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL, JavaScript

Backend & Data: Flask, REST APIs, Data Pipelines, MySQL, Data Validation & Integrity

Machine Learning & Statistics: Supervised & Unsupervised Learning, Linear & Logistic Regression, SVM, Random Forest, Decision Trees, Naive Bayes, KNN, K-Means Clustering, Regularization (Ridge & Lasso), Data Preprocessing, Pandas & NumPy

Data Visualization: Matplotlib, Seaborn, Chart.js, Apache Superset (Familiar)

Tools & Platforms: Git, GitHub, GitLab, CI/CD (basic), Firebase, Modular Architecture, Advanced Excel (Pivot Tables, VLOOKUP, Data Cleaning)

Web Technologies: HTML5, CSS3, Tailwind CSS, Streamlit

EDUCATION

B.Tech in Computer Science and Engineering — Adamas University, Kolkata | 2023 – 2027
CGPA: **8.65**

Class XII (CBSE) — Kalyani Public School | 2023
Score: **83.4%**

Class X (CBSE) — Adamas World School | 2021
Score: **93.8%**

EXPERIENCE

Artificial Intelligence Trainee — Infosys Springboard | Oct 2025 – Dec 2025

- Implemented Python-based AI and ML workflows covering search algorithms, heuristic problem-solving, and data preprocessing pipelines, strengthening foundations in algorithmic thinking and scalable logic design.
- Applied structured data handling and modular workflow design to build end-to-end ML pipelines, reinforcing understanding of feature engineering, model evaluation, and data integrity validation.
- Collaborated on code reviews and version control workflows using Git, applying software engineering best practices to maintain code quality and system reliability across iterative development cycles.

Software Development Intern — *Harit Vikas Technologies Pvt. Ltd. | Jan 2026 – Present*

Saaya MRV Platform (Ongoing Opensource Project)

- Refactored the platform's core architecture to introduce a modular, extensible component model, reducing feature integration overhead and improving maintainability across a 15+ module codebase
- Designed and implemented a unified Source Adapter layer abstracting data ingestion from heterogeneous sources — Google Firestore, REST APIs, and GeoTIFF satellite imagery — enforcing a consistent data validation interface and reducing new-source integration time by ~60%.
- Engineered a geospatial batch-placement algorithm to auto-generate and persist 500+ location markers from coordinate datasets, reducing manual data entry overhead by 90% and eliminating user-input errors through schema-level data validation.
- Built a satellite-driven NDVI data pipeline to ingest, process, and visualize plant health time-series data, enabling scalable environmental monitoring across multiple project sites.
- Integrated bulk data ingestion via Excel/CSV upload and implemented cluster-level delete operations, enabling users to manage thousands of geospatial data points with enforced data integrity constraints.
- Improved platform reliability by diagnosing and resolving critical dependency conflicts and implementing regression-safe update practices, ensuring system stability across iterative releases.

Research Intern — *IEEE SMC Student Branch Chapter, KGEC (In collaboration with IEEE SMC Kolkata Chapter & IEEE Kolkata Section) | May 2026 – August 2026*

- Selected for the highly competitive Data Science and Intelligent Analytics track (Rank 16/80; 21.05% acceptance rate from 380+ applicants nationwide); conducting research in data-driven pattern analysis and intelligent systems under IEEE SMC supervision.

PROJECTS

AI-Based Research Paper Discovery System with Query Disambiguation (Ongoing)

- Developing a scalable NLP-based information retrieval system to query, rank, and surface scientific research papers from open-access sources (arXiv API), ordered by recency and relevance score
- Implementing text preprocessing and concept extraction using SpaCy and regex
- Designing an abbreviation disambiguation module to resolve ambiguous queries (e.g., "POP") via user-guided selection
- Building a context-aware search pipeline with query-to-concept mapping and NLP-driven concept extraction, enabling low-latency resolution of ambiguous user queries through an abbreviation disambiguation module
- Tech: Python, SpaCy, Regex, arXiv API

SpotNFix — Crowdsourced Civic Issue Reporting Platform

- Built a web-based system for reporting civic issues using HTML, CSS, JavaScript, and Firebase.
- Integrated Google Maps API for auto-location and issue visualization.
- Implemented an image classification pipeline by integrating pre-trained image classification endpoints for automated content verification., duplicate detection, and issue categorization — reducing invalid report submissions and improving data integrity of the crowdsourced dataset.

CERTIFICATIONS

- C Programming — Great Learning Academy
- Fundamentals of Python Programming — Infosys
- The Joy of Computing with Python — NPTEL
- Artificial Intelligence: Search Methods for Problem Solving — NPTEL

ACHIEVEMENTS

- 5 star Gold Badge in C Programming on HackerRank
- Qualifier — Smart India Hackathon (Internal) 2025
- Clash of Coders — 2nd Round Qualifier